Structural CAD Standard for Automated Revit Modelling

Project	J-1153, Reis Magos Goa Housing
Issued by	MOFA Studios Pvt. Ltd.
For	The structural consultant
File audited	`1153 - 1ST FLOOR PLAN ( C WING ) 14-07-2026.dwg`
Version	1.0, 25 September 2026

1. What this document is for

We generate the structural Revit model from your CAD drawing instead of redrawing it by hand. A script reads the DWG layers, matches each shape to its mark, joins the mark to the schedule row, and places the Revit element. Anything the script cannot read with certainty is not guessed; it comes back to you as a written query.

This only works if the drawing is consistent in a few specific ways. This document sets out those ways.

What we are not asking for. We are not asking you to model in Revit, to change how the drawing looks when printed, to redraw existing work, or to adopt a new CAD package. Every rule below is about how entities are placed on layers in the DWG you already produce.

What changes for you. Roughly six drafting habits, listed in section 5, plus three schedules delivered as Excel alongside the DWG. Nothing in section 5 takes longer to draw than what is being drawn today.

Why it is worth it. On the audited file, 4,200 entities carry the whole first floor of one wing. Reading them automatically takes minutes. Redrawing them takes days, and takes days again at every reissue.

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2. Status of the current file

We read `1153 - 1ST FLOOR PLAN ( C WING ) 14-07-2026.dwg` in full. Appendix A lists what we found, with counts. In summary:

Already correct and requiring no change

- AutoCAD 2010 format, `INSUNITS` = 4 (millimetres), `MEASUREMENT` = 1 (metric), drawn 1:1.
- Columns, beams, girders, floating columns and the sunk zones are already on their own layers.
- Column outlines are real closed rectangles: de-rotated, they give exact schedule sizes (230 x 950, 230 x 1200, 350 x 750 and so on).
- Beams are multilines at a true 230 width, girders at 350.

- All marks are live text, not exploded geometry, at consistent heights (250 mm on columns and beams, 300 mm on slabs).
- No external references. Nothing is xref'd in, so the file is self-contained.
- The plan is orthogonal to itself: beam multilines run at 2 degrees and 92 degrees with no drift.

Blocking automation today

#	What we found	Section that fixes it
1	The `SLAB` layer holds 6 open fragments. There is no closed outline anywhere to build a Revit floor from, against 65 slab tags.	5.5
2	Beam multilines use **top** justification, so every drawn line is a beam face, not a centreline: 115 mm off for beams, 175 mm for girders.	5.3
3	There is no grid. No grid lines, no bubbles, no grid layer anywhere in the file.	4.5
4	The plan sits about 2.9 km from the drawing origin with no point agreed against the architectural model.	4.4
5	All three schedules are loose text, not tables. The beam schedule alone is 1,934 separate text objects.	6
6	Sunk layers are named `300mm SUNK` and `900MM SUNK`; the legend on the same sheet calls for 100, 150 and 400 sunk.	4.6, 5.6
7	375 entities sit at a Z other than zero, and the drawing extents report a Z of 30,544,084 mm.	4.3

3. Deliverables, per floor, per issue

#	Deliverable	Format
1	Structural plan, one floor per file	DWG, AutoCAD 2010 or later
2	Column schedule	XLSX, one row per mark
3	Beam and girder schedule	XLSX, one row per mark
4	Slab schedule	XLSX, one row per mark
5	Level table, plinth to terrace	XLSX or PDF
6	Sections through sunk areas, pool, deck, stairs and lift	PDF
7	Revision note listing what changed since the previous issue	PDF or in the DWG title block

File naming

```
J1153_<Wing>_<Level>_STR_<YYYY-MM-DD>.dwg
J1153_<Wing>_<Level>_STR_<YYYY-MM-DD>_COL.xlsx
```

```
J1153_<Wing>_<Level>_STR_<YYYY-MM-DD>_BEAM.xlsx
J1153_<Wing>_<Level>_STR_<YYYY-MM-DD>_SLAB.xlsx
```

Example: `J1153_C_01_STR_2026-07-14.dwg`

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4. Drawing setup

4.1 Units and scale

Millimetres, drawn 1:1. `INSUNITS` = 4, `MEASUREMENT` = 1. Nothing scaled, stretched or exploded. *The current file already complies.*

4.2 One floor per file

Model space holds the structural plan for one floor only. Schedules, legends, details, title block and key plan may sit beside the plan in model space, but on their own layers (section 6) so they can be ignored by the reader.

4.3 Flat at zero

Every entity at Z = 0. Run `FLATTEN` before issue, then check that `EXTMIN` and `EXTMAX` both report Z = 0.

In the audited file: 375 entities carry a Z other than zero — 244 of them at Z = −229, and others at −11,425, −11,197, −8,962, +7,506, +14,784. Drawing extents report a Z range of −22,493 to 30,544,084 mm. A script reading plan coordinates has to discard Z entirely, which also means it cannot trust any Z it does find.

4.4 Origin and setting out

- One base point, shared with the architectural model, agreed once for the whole project.
- The same XY position for every floor and every wing. A floor must sit directly above the floor below with no shift.
- Never re-rotate the plan between issues. The current 2 degree rotation is fine and can stay; it simply must not change.
- Mark the shared base point with a single point or small cross on layer `S-REF`.

In the audited file: the plan occupies about 25.4 m x 56.3 m, positioned roughly 2.9 km from the drawing origin, with nothing identifying a shared point.

4.5 Grid

The structural grid goes on `S-GRID`: grid lines as lines, bubbles as circles with the grid label as text inside. If the structural grid is the architect's grid, copy it in from the architectural file rather than redrawing it.

In the audited file: there is no grid of any kind. Columns are numbered 1 to 32 with no lines to locate them against. There are 34 dimensions on the `DIM` layer, but with no grid there is nothing for them to dimension from.

4.6 Housekeeping before issue

- `OVERKILL` run, so no entity is drawn twice on top of another.
- `PURGE` run, so empty layers, unused blocks and unused styles are gone.
- `AUDIT` run, with errors fixed.
- No structural element inside a block. Blocks are fine for title blocks, logos, section marks and stair symbols; not for columns, beams, slabs or walls.
- Layer names spelled one way only.

In the audited file: 10 of 51 layers carry no entities; `RCC PARDI` and `R.C.C. PARDI` both exist as layers with only the second one used; 2 column outlines are exact duplicates lying on top of each other. 69 block definitions exist, none of which contain structural elements, which is correct.

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5. How each element is drawn

5.1 Columns

- One **closed** rectangle per column, drawn at the scheduled size, on `S-COL`.
- Columns that stop at this floor go on `S-COL-STOP` instead, so we know not to carry them up.
- Floating columns go on `S-COL-FLOAT`.
- Nothing else on these layers. No hatch, no text, no detail linework.

In the audited file: 68 outlines on `COLUMN`, of which 61 are closed rectangles that de-rotate to exact schedule sizes, 2 are closed 17-point outlines (the lift walls), and 5 are **open** 5-point polylines that a script cannot fill. A further 8 rectangles of 495 x 1141 sit on the `COLUMN` layer at 0 degrees, a size that appears nowhere in the column schedule; please confirm what these are. The 6 outlines on `FL COL` are all correctly closed.

5.2 Walls

Lift walls (LW-1 to LW-4) and the RCC pardi walls go on `S-WALL` as closed outlines, with their thickness as drawn. They are not columns and must not sit on the column layer.

In the audited file: the lift wall pieces are drawn on `COLUMN` as 230 x 2110 and 230 x 1950 rectangles and as two 17-point outlines. The RCC pardi is on `R.C.C. PARDI`, one closed and one open outline.

5.3 Beams and girders — the most important rule

- **One centreline per span**, on `S-BEAM` for beams and `S-GIRDER` for girders.
- If drawn as a multiline, set justification to **Zero**, not Top or Bottom.
- One entity type throughout. Pick multiline or polyline and stay with it.
- Girders drawn at the scheduled width, not at a single nominal width for all of them.
- Every line parallel to the grid, or at the true angle of that part of the building.

In the audited file:

Finding	Count
Beam multilines, width 230, justification **TOP**	113
Girder multilines, width 350, justification **TOP**	2
Beams drawn as plain lines instead	49
Beams drawn as open polylines instead	8
Girders drawn as plain lines	12
Girder tags on the plan, against 2 girder multilines	9

Top justification means the line in the file lies on one face of the beam, 115 mm from its centre (175 mm for a 350 girder). A script can shift a line by 115 mm, but nothing in the file says which side to shift it to, and guessing wrong puts a beam a full width away from its columns.

Beam multilines are clean on angle: 59 at 92 degrees and 56 at 2 degrees. The plain lines are not: 13 run at 69 degrees, which is the angled wing and correct, but 2 run at 0 degrees. On `GIRDER BEAM`, 2 lines run at 0 degrees, 2 at 85 and 2 at 6, where 2 or 92 is expected.

5.4 Slabs

- **One closed outline per slab panel**, on `S-SLAB`.
- Edges land on the beam centrelines, or on the slab edge where there is no beam.
- One tag per panel, even where several panels share the same slab type.

In the audited file: the `SLAB` layer holds 6 open polylines of 3 and 4 points. There is nothing closed to convert into a Revit floor. The 65 slab tags cover 19 unique marks, so several panels share a mark: S1 is used 14 times, S3 12 times, S4 8 times, S8 6 times. Each of those placements needs its own outline.

5.5 Sunk areas, openings and stairs

- A **closed outline** for each sunk zone on `S-SUNK-100`, `S-SUNK-150` or `S-SUNK-400`, named for its depth in millimetres. Add layers for other depths as the same pattern.
- Pool and deck slabs on `S-SUNK-POOL` and `S-SUNK-DECK`.
- A hatch may sit on top for printing, but the closed outline must exist. A hatch alone is not enough.
- Openings and cut-outs as closed outlines on `S-OPEN`.
- Stairs and landings as closed outlines on `S-STAIR`.

In the audited file: sunk zones exist only as hatches, 12 on `300mm SUNK` (pattern ANSI33) and 7 on `900MM SUNK` (pattern FLEX), with 1 more on `SUNK` and 10 on `HATCH`. The legend on the same sheet lists 100MM SUNK, 150MM SUNK, 400MM SUNK, POOL SLAB and DECK SLAB. The layer names and the legend do not agree, and we cannot tell which hatch is which depth.

5.6 What must not be mixed

One layer, one kind of thing. Do not place tags, hatches, dimensions or detail linework on an element layer, and do not place two element types on one layer.

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6. Layer standard

Element geometry and its tag are always on separate layers, so the reader can pick up shapes and text independently.

Layer	Contents	Entity type
`S-COL`	Columns continuing above this floor	Closed polyline
`S-COL-STOP`	Columns stopping at this floor	Closed polyline
`S-COL-FLOAT`	Floating columns	Closed polyline
`S-WALL`	Lift walls, RCC pardi, retaining walls	Closed polyline
`S-BEAM`	Beam centrelines	Line or polyline
`S-GIRDER`	Girder centrelines	Line or polyline
`S-SLAB`	One closed outline per slab panel	Closed polyline
`S-SUNK-100`	Sunk 100 zones	Closed polyline
`S-SUNK-150`	Sunk 150 zones	Closed polyline
`S-SUNK-400`	Sunk 400 zones	Closed polyline
`S-SUNK-POOL`	Pool slab	Closed polyline
`S-SUNK-DECK`	Deck slab	Closed polyline
`S-OPEN`	Openings and cut-outs	Closed polyline
`S-STAIR`	Stair flights and landings	Closed polyline
`S-GRID`	Grid lines and bubbles	Line, circle, text
`S-REF`	Shared base point, level marks	Point, text
`S-COL-TAG`	Column marks	Text
`S-WALL-TAG`	Wall marks	Text
`S-BEAM-TAG`	Beam marks	Text
`S-GIRDER-TAG`	Girder marks	Text
`S-SLAB-TAG`	Slab marks	Text
`S-HATCH`	Hatches for printing only	Hatch
`S-DIM`	Dimensions	Dimension
`S-ANNO`	Notes, legend, schedules, title block, details	Any

If you already run a layer standard of your own, send it to us. Mapping our reader onto your names is a ten-minute job and is preferable to you changing how you draft. What matters is that the mapping is one-to-one

and does not change between issues.

Mapping from the current file

Current layer	Becomes
`COLUMN`	`S-COL`, with the lift wall pieces moved to `S-WALL`
`FL COL`	`S-COL-FLOAT`
`BEAM`	`S-BEAM`
`GIRDER BEAM`	`S-GIRDER`
`SLAB`	`S-SLAB`
`R.C.C. PARDI`, `RCC PARDI`	`S-WALL` (one layer, not two)
`300mm SUNK`, `900MM SUNK`, `SUNK`, `HATCH`	`S-SUNK-<depth>`, named for the real depth
`colno`	`S-COL-TAG`
`BEAM NO (H)`, `BEAM NO (V)`	`S-BEAM-TAG` (one layer, not two)
`SLAB NO`	`S-SLAB-TAG`
`COL HATCH`, `GIRDER-HATCH`, `RET WALL HATCH`	`S-HATCH`
`DIM`	`S-DIM`
`SHEET`, `MAIN TEXT`, `Col. sch.`, `beam sch. text`, `slab sch. text`, `beam & slab detail`	`S-ANNO`

7. Marks and tags

7.1 Format

Element	Prefix	Example
Column	`C-`	`C-12`
Floating column	`FC-`	`FC-3`
Lift wall	`LW-`	`LW-2`
Beam	`B-`	`B-34`
Cantilever beam	`BC-`	`BC-6`
Plinth beam	`BP-`	`BP-1`
Mid-landing beam	`MLB-`	`MLB-1`
Girder beam	`GB-`	`GB-5`
Slab	`S-`	`S-4A`
Staircase slab	`ST-`	`ST-2A`

7.2 Placement

1. **One mark per text object.** A text object reading `B107/BP` or `/B60` describes two elements and cannot be split reliably. Use two text objects.
2. **Within 300 mm** of the element it labels, or inside it. Where a leader is needed for clarity in print, keep the text within 300 mm and let the leader run the other way.
3. **One tag per element**, once. No duplicated tags, no tag without an element, no element without a tag.
4. Real text or mtext, never exploded to lines.

In the audited file: tag placement is the single most uneven area.

	Column tags	Beam tags
Tags found	48	123
Nearest element, minimum	191 mm	22 mm
Nearest element, median	593 mm	90 mm
Nearest element, maximum	1,446 mm	2,705 mm
More than 300 mm away	41 of 48	27 of 123
More than 1,000 mm away	3 of 48	4 of 123

Beam tagging is mostly good — the median beam tag is 90 mm from its beam. The outliers are the problem: B87 at 2,705 mm, B54 at 2,225 mm, BP3 at 1,299 mm and BP2 at 1,279 mm are closer to a different beam than to their own.

Column tagging fails the rule almost everywhere, with a median of 593 mm. There are also 48 tags against 74 column and floating-column outlines, so 26 outlines carry no mark at all.

Compound marks currently in use that need splitting: `/B60`, `B33/BP`, `B35/BP`, `B91/BP`, `B107/BP`, `BC6/BP`.

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8. Schedules as data

Send the column, beam and slab schedules as Excel, in addition to drawing them on the sheet. The drawn schedule stays exactly as it is for printing; the Excel is a parallel delivery, not a replacement.

Rules

1. One row per mark. Grouped rows are split before delivery: `B-2,4,7 | 230x600` becomes three rows.
2. One value per cell. No units inside the cell, no ranges, no "as above", no merged cells.
3. A header row with stable column names that do not change between issues.
4. Every row complete. A blank depth or an unreadable size stops that element being modelled.

Use the workbook issued with this document. `J1153_C_01_STR_2026-07-14.xlsx` carries one sheet per element type with the columns already set out, the marks we read from your drawing already entered in the Mark column, and dropdowns on the fields that must match a fixed list. Its Read Me sheet repeats these rules.

The columns are, per sheet, with the required ones marked *:

Sheet	Columns
Levels	Level Name* / Structural Level SSL (mm)* / Floor to Floor (mm) / Datum Note
Columns	Mark* / CAD tag / Type* / Width (mm) */ Depth (mm)* / Base Level* / Top Level* / Concrete Grade* / Main Steel / Links / Continues Above / Remark
Beams	Mark* / CAD tag / Type* / Width (mm) */ Depth (mm)* / Level* / Top of Beam Offset (mm) / Concrete Grade* / Top Steel / Bottom Steel / Extra Top / Stirrups / Sunk Depth (mm) / Remark
Slabs	Mark* / CAD tag / Thickness (mm) */ Level* / Top Offset (mm) / Concrete Grade* / Span Type / Sunk Depth (mm) / Top Steel / Bottom Steel / Remark
Walls	Mark* / CAD tag / Thickness (mm) */ Base Level* / Top Level* / Concrete Grade* / Vertical Steel / Horizontal Steel / Remark
Sunk and Openings	ID* / Type* / Depth (mm) */ Level* / In Slab Panel / Where / Remark
Stairs	Mark* / CAD tag / Waist (mm) */ Level From* / Level To* / Concrete Grade* / Main Steel / Remark
Revisions	Rev* / Date* / Issued By / What Changed* / Marks Affected

The **CAD tag** column shows the tag exactly as it appears in your drawing, so a mark that reads differently in the schedule and on the plan is visible at a glance. Correct it if we have read it wrongly.

The **Level** columns are the one addition to what is drawn on the sheet today. A Revit element has to sit on a level, and the drawn schedules carry no level information at all.

In the audited file: all three schedules exist only as free text placed on the sheet — 1,934 separate text objects for the beam schedule, 230 for the column schedule and 161 for the slab schedule. Rows can only be rebuilt by guessing which texts line up horizontally, which is not reliable enough to model from. Grouped entries such as `2,4,7`, `10,14,22,26`, `107,109,111` and `55,56,58,79` are already in use and each needs splitting.

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9. Levels

Send a level table for the wing, once, and reissue it only when it changes.

Column	Meaning
`Level Name`	Plinth, 1st Floor, 2nd Floor, Terrace
`Structural Level (mm)`	Top of slab, against project datum
`Floor to Floor (mm)`	To the level above
`Datum Note`	What the project zero refers to

Where a beam or slab does not sit at the floor level — a sunk beam, a stepped beam, a partly sunk beam such as B-34 or GB-5 — give the drop in the schedule's `Sunk` column, and give one section through it.

In the audited file: there is no level information in the DWG. The `S- LEVEL` layer exists but is empty.

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10. Submission checklist

To be confirmed before each issue.

	Check
1	Units are millimetres, drawing is 1:1, nothing exploded
2	`FLATTEN` run; `EXTMIN` and `EXTMAX` both report Z = 0
3	Plan sits on the agreed base point, same XY as the floor below, rotation unchanged
4	`S-GRID` present, matching the architectural grid
5	Every column, wall, slab, sunk zone, opening and stair outline is **closed**
6	Every beam and girder is a single centreline, justification Zero
7	Girders drawn at their scheduled width
8	One closed outline per slab panel, including repeated panels
9	Sunk layers named for their real depth, matching the legend
10	One mark per text object, within 300 mm of its element, one tag per element
11	Every mark on the plan has a schedule row, and every schedule row has a mark on the plan
12	`OVERKILL`, `PURGE` and `AUDIT` run; no duplicated entities; no empty layers
13	Column, beam and slab schedules attached as Excel, one row per mark, no blanks
14	Level table attached or confirmed unchanged
15	Revision note listing what changed since the previous issue

Appendix A — Audit of the 14-07-2026 Wing C file

Read from the DWG itself, converted to DXF without altering the original. Raw output is in `Wing-C-Audit-Evidence.txt` in this folder.

File

Item	Value
Format	AutoCAD 2010 (AC1024)
`INSUNITS` / `MEASUREMENT`	4 (millimetres) / 1 (metric)
Entities	4,202 — 4,200 in model space, 2 in paper space
Layers	51 defined, 10 of them empty
Block definitions	69, no external references
Text styles	7; 2,772 of 2,873 texts use `arial`
Plan extent	about 25.4 m x 56.3 m
Distance from origin	about 2.9 km
Plan rotation	2.0 degrees
Entities with Z not zero	375
Reported Z extents	−22,493 to 30,544,084 mm

Columns

Item	Value
Outlines on `COLUMN`	68: 61 closed rectangles, 2 closed 17-point, 5 open
Outlines on `FL COL`	6, all closed
Exact duplicate outlines	2
Rotation	53 rectangles at 2 degrees, 8 at 0 degrees
Sizes recovered	230x950 (11), 230x2110 (8), 230x1050 (5), 230x900 (5), 230x1200 (5), 350x750 (5), 230x1950 (4), 230x1500 (4), 230x1350, 230x750, 230x230, 300x1050, 350x1480, 350x2000
Unexplained	8 rectangles of 495 x 1141, at 0 degrees, matching no schedule size
Tags	48, against 74 outlines

Beams and girders

Item	Value
`BEAM`	113 multilines (width 230, justification TOP), 49 lines, 8 open polylines, 3 hatches
`GIRDER BEAM`	2 multilines (width 350, justification TOP), 12 lines, 4 polylines
Multiline angles	92 degrees (59), 2 degrees (56)
Beam line angles	2 (17), 92 (17), 69 (13), 0 (2)
Girder line angles	92 (6), 0 (2), 85 (2), 6 (2)
Tags	123 on two layers, 122 unique marks

Slabs

Item	Value
`SLAB`	6 open polylines, no closed outline
`SLAB NO`	65 tags, 19 unique marks
Most repeated marks	S1 (14), S3 (12), S4 (8), S8 (6)
`SLAB MARK`	95 open 4-point polylines, tag boxes
Sunk hatches	`300mm SUNK` 12 (ANSI33), `900MM SUNK` 7 (FLEX), `SUNK` 1, `HATCH` 10
Legend calls for	100MM SUNK, 150MM SUNK, 400MM SUNK, POOL SLAB, DECK SLAB
Slab thicknesses in the schedule	125, 150, 175, 200

Schedules and setting out

Item	Value
Beam schedule	1,934 text objects, 142 unique strings
Column schedule	230 text objects
Slab schedule	161 text objects
Dimensions	34
Grid	none — no grid lines, bubbles or grid layer
Level information	none; `S- LEVEL` layer is empty

Appendix B — Queries on the current drawing

These are needed before Wing C first floor can be modelled. Please answer against the mark, not in general terms.

#	Query
1	Should beam centrelines run column face to face, or column centre to column centre? We will follow whichever you use, but it must be the same on every floor.
2	Which hatch is 100, 150 and 400 sunk? The layers are named `300mm SUNK` and `900MM SUNK`, the legend lists 100, 150 and 400.
3	Marks 11, 15 and 19 appear on the plan. We could not find a matching row in the column schedule. Are they column stops, or is the row missing?
4	Eight rectangles of 495 x 1141 sit on the `COLUMN` layer at 0 degrees. What are they?
5	C-30 is drawn 230 x 827 but scheduled 230 x 950, with an untagged 230 x 950 outline 0.9 m away. Which is correct?
6	26 column and floating-column outlines carry no mark. Please tag them.
7	B-103 and B-106 carry no depth. GB-5 reads `1480 800/`. B-34 reads `700/` and is noted as partly sunk. Please complete these rows and give the step.
8	B-60 appears on the plan only inside the compound text `/B60`. B-94 and B-97 are scheduled but we found no tag. GB-3A is tagged twice.
9	Floor-to-floor heights and structural levels from plinth to terrace.
10	Which point do we share with the architectural model as the common origin?
11	Girder widths run from 350 to 1500 in the schedule but only two girders are drawn with a width. Confirm we take girder widths from the schedule until 5.3 is applied.
12	Grade is M30 for beams and slabs and M40 for girders. Confirm this holds for every floor, or tell us where it changes.

Appendix C — Contact

Queries on this document to MOFA Studios Pvt. Ltd. We are happy to run the checklist in section 10 against a trial file before you adopt it, at no cost to your programme, so that any disagreement surfaces on one floor rather than on the whole project.